

Table A.1

Descriptive statistics of the model variables for rust

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	348.79	304.00	270.09	(346.85, 350.74)
<i>ns</i>	1.03	1.00	0.21	(1.03, 1.04)
<i>nf</i>	6.43	2.00	92.97	(5.77, 7.11)
<i>rexp</i>	74.79	1.13	1321.50	(65.27, 84.32)
<i>sexp</i>	3835.32	673.5	6940.66	(3785.27, 3885.37)
<i>entropy</i>	0.8812	0.2422	1.2184	(0.87, 0.89)
<i>days_to_first_fix</i>	3.46	0.00	47.83	(3.12, 3.81)
<i>exp</i>	4255.27	889.25	7357.31	(4202.22, 4308.33)

Table A.2

Descriptive statistics of the model variables for Tensorflow

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	268.72	285.00	140.67	(267.65, 269.81)
<i>ns</i>	1.07	1.00	0.36	(1.07, 1.08)
<i>nf</i>	5.23	2.00	35.58	(4.96, 5.51)
<i>rexp</i>	17.61	1.13	246.65	(15.72, 19.51)
<i>sexp</i>	20180.28	596.5	34515.36	(19915.01, 20445.56)
<i>entropy</i>	0.8850	0.6665	1.0911	(0.88, 0.89)
<i>days_to_first_fix</i>	24.60	0.00	87.73	(23.93, 25.28)
<i>exp</i>	21150.18	780.75	35330.65	(20878.64, 21421.73)

A. Dataset Description

A heat map is a data visualisation that represents different data intensities within a dataset using colour gradients. It is often used to identify correlations, patterns, and trends in data, making it easier to display complex information quickly. It conveys potential relationships between the variables shown on the x- and y-axes by using different colors, or varying shades of the same color, to indicate various values. Darker hues or shades typically indicate larger or more quantities of the values shown in the heat map. The reddish color represents the positive correlation, and the bluish color represents the negative correlation among the variables. We will also assess distribution shapes, variability, and central tendencies using descriptive statistics. It also enable us to identify outliers and unusual values. In the below sections, we have visualized each software projects using correlation metrics and descriptive statistics.

A.1. Rust

The heat map of the correlation coefficient among the variables of our interest for the Rust data set is shown in Fig. A.1.

Table A.1 shows the statistics that can help to understand the distribution and variability of the different variables in the Rust dataset.

A.2. TensorFlow

The heat map of the correlation coefficient among the variables of our interest for the Tensorflow data set is shown in Fig. A.2.

Table A.2 shows the statistics that can help to understand the distribution and variability of the different variables in the Tensorflow dataset.

A.3. Rails

The heat map of the correlation coefficient among the variables of our interest for the Rails data set is shown in Fig. A.3.

Table A.3 shows the statistics that can help to understand the distribution and variability of the different variables in the Rails dataset.

Table A.3

Descriptive statistics of the model variables for Rails

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	252.61	232.00	169.90	(251.23, 254.00)
<i>ns</i>	1.15	1.00	0.66	(1.15, 1.16)
<i>nf</i>	2.76	1.00	8.20	(2.70, 2.84)
<i>rexp</i>	65.09	1.119	549.86	(60.62, 69.57)
<i>sexp</i>	428.39	77.0	746.29	(422.32, 434.48)
<i>entropy</i>	0.6316	0.00	0.8885	(0.63, 0.64)
<i>days_to_first_fix</i>	106.44	0.00	352.34	(103.57, 109.31)
<i>exp</i>	1713.74	450.0	2613.08	(1692.47, 1735.02)

Table A.4

Descriptive statistics of the model variables for VSCode

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	60.60	63.00	27.40	(60.37, 60.84)
<i>ns</i>	1.05	1.00	0.26	(1.05, 1.05)
<i>nf</i>	4.59	1.00	55.96	(4.11, 5.07)
<i>rexp</i>	52.53	1.40	422.99	(48.89, 56.17)
<i>sexp</i>	4739.20	2312.5	5942.17	(4688.06, 4790.36)
<i>entropy</i>	0.7228	0.00	1.0209	(0.71, 0.73)
<i>days_to_first_fix</i>	2.67	0.00	32.59	(2.40, 2.96)
<i>exp</i>	6601.52	4540.75	6804.46	(6542.96, 6660.10)

Table A.5

Descriptive statistics of the model variables for WP-Calypso

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	138.94	146.00	75.89	(138.11, 139.79)
<i>ns</i>	1.11	1.00	0.39	(1.11, 1.12)
<i>nf</i>	3.85	2.00	27.52	(3.55, 4.16)
<i>rexp</i>	48.91	1.11	337.92	(45.16, 52.66)
<i>sexp</i>	507.74	206.0	778.86	(499.11, 516.39)
<i>entropy</i>	0.77	0.4262	0.99	(0.76, 0.79)
<i>days_to_first_fix</i>	45.27	0.00	154.18	(43.56, 46.98)
<i>exp</i>	701.74	337.0	998.26	(690.67, 712.83)

A.4. VSCode

The heat map of the correlation coefficient among the variables of our interest for the VSCode data set is shown in Fig. A.4.

Table A.4 shows the statistics that can help to understand the distribution and variability of the different variables in the VSCode dataset.

A.5. WP-Calypso

The heat map of the correlation coefficient among the variables of our interest for the WP-Calypso data set is shown in Fig. A.5.

Table A.5 shows the statistics that can help to understand the distribution and variability of the different variables in the WP-Calypso dataset.

A.6. Camel

The heat map of the correlation coefficient among the variables of our interest for the Camel data set is shown in Fig. A.6.

Table A.6 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A.6

Descriptive statistics of the model variables for Camel

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	47.31	31.00	44.80	(46.81, 47.82)
<i>ns</i>	1.28	1.00	0.83	(1.27, 1.28)
<i>nf</i>	6.77	2.00	41.32	(6.30, 7.24)
<i>rexp</i>	63.45	1.06	682.88	(55.75, 71.15)
<i>sexp</i>	5149.49	844.00	8191.84	(5057.15, 5241.85)
<i>entropy</i>	1.0853	0.8136	1.3254	(1.07, 1.10)
<i>days_to_first_fix</i>	48.74	0.00	225.34	(46.20, 51.28)
<i>exp</i>	15312.83	5702.50	19512.00	(15092.86, 15532.80)

Table A.7

Descriptive statistics of the model variables for CoreFX

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	193.68	188.00	109.81	(192.37, 195.01)
<i>ns</i>	1.13	1.00	0.41	(1.13, 1.14)
<i>nf</i>	11.71	2.00	78.34	(10.77, 12.66)
<i>rexp</i>	25.11	1.07	228.37	(22.37, 27.85)
<i>sexp</i>	3538.41	440.00	8407.69	(3437.42, 3639.41)
<i>entropy</i>	1.17	0.83	1.52	(1.16, 1.19)
<i>days_to_first_fix</i>	13.53	0.00	85.95	((12.50, 14.57)
<i>exp</i>	4613.17	788.00	11077.52	(4480.12, 4746.24)

Table A.8

Descriptive statistics of the model variables for Django

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	77.25	35.00	81.19	(76.27, 78.23)
<i>ns</i>	1.45	1.00	0.72	(1.45, 1.46)
<i>nf</i>	4.37	2.00	33.82	(3.97, 4.79)
<i>rexp</i>	41.75	1.03	414.70	(36.74, 46.76)
<i>sexp</i>	802.20	268.00	1432.75	(784.91, 819.50)
<i>entropy</i>	0.7102	0.00	1.0183	(0.70, 0.72)
<i>days_to_first_fix</i>	153.80	0.00	377.60	(149.25, 158.36)
<i>exp</i>	2335.92	836.25	3360.53	(2295.35, 2376.49)

A.7. CoreFX

The heat map of the correlation coefficients among the variables of interest in the CoreFX dataset is shown in Fig. A.7.

Table A.7 shows the statistics that can help to understand the distribution and variability of the different variables in the CoreFX dataset.

A.8. Django

The heat map of the correlation coefficients among the variables of interest in the Django dataset is shown in Fig. A.8.

Table A.8 shows the statistics that can help to understand the distribution and variability of the different variables in the Django dataset.

A.9. Nova

The heat map of the correlation coefficients among the variables of interest in the Nova dataset is shown in Fig. A.9.

Table A.9 shows the statistics that can help to understand the distribution and variability of the different variables in the Nova dataset.

Table A.9

Descriptive statistics of the model variables for Nova

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	168.82	149.00	118.28	(167.40, 170.25)
<i>ns</i>	1.14	1.00	0.44	(1.14, 1.15)
<i>nf</i>	4.49	2.00	36.56	(4.05, 4.94)
<i>rexp</i>	124.65	1.04	2619.02	(93.01, 156.30)
<i>sexp</i>	363.62	90.0	744.30	(354.63, 372.62)
<i>entropy</i>	0.8764	0.7219	1.0609	(0.86, 0.89)
<i>days_to_first_fix</i>	92.37	0.00	220.29	(89.71, 95.04)
<i>exp</i>	532.47	164.5	980.44	(520.62, 544.32)

Table A.10

Descriptive statistics of the model variables for JGroup

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	10.83	11.00	7.07	(10.73, 10.94)
<i>ns</i>	1.17	1.00	0.48	(1.17, 1.18)
<i>nf</i>	2.51	1.00	13.09	(2.32, 2.70)
<i>rexp</i>	47.87	1.0039	1232.16	(29.88, 65.88)
<i>sexp</i>	3342.80	2539.0	3167.17	(3296.54, 3389.07)
<i>entropy</i>	0.3983	0.0	0.8194	(0.39, 0.41)
<i>days_to_first_fix</i>	73.14	0.00	312.84	(68.58, 77.72)
<i>exp</i>	8293.86	7846.0	6150.00	(8204.03, 8383.71)

Table A.11

Descriptive statistics of the model variables for Fabric 8

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	32.10	31.00	16.72	(31.82, 32.40)
<i>ns</i>	1.53	1.00	1.79	(1.51, 1.57)
<i>nf</i>	9.46	2.00	84.56	(7.98, 10.95)
<i>rexp</i>	186.125	1.20	3249.38	(129.14, 243.10)
<i>sexp</i>	895.08	179.00	1469.96	(869.31, 920.86)
<i>entropy</i>	1.0616	0.2108	1.5323	(1.03, 1.09)
<i>days_to_first_fix</i>	25.58	0.00	97.95	(23.86, 27.30)
<i>exp</i>	6918.40	3954.00	8384.15	(6771.39, 7065.43)

A.10. JGroup

The heat map of the correlation coefficients among the variables of interest in the JGroup data set is shown in Fig. A.10.

Table A.10 shows the statistics that can help to understand the distribution and variability of the different variables in the JGroup dataset.

A.11. Fabric 8

The heat map of the correlation coefficient among the variables of our interest for the fabric data set is shown in Fig. A.11.

Table A.11 shows the statistics that can help to understand the distribution and variability of the different variables in the fabric dataset.

A.12. Broadleaf

The heat map of the correlation coefficient among the variables of our interest for the Broadleaf data set is shown in Fig. A.12.

Table A.12 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A.12

Descriptive statistics of the model variables for Broadleaf

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	21.07	24.00	11.00	(20.88, 21.27)
<i>ns</i>	1.33	1.00	0.96	(1.32, 1.35)
<i>nf</i>	9.77	2.00	102.34	(7.97, 11.58)
<i>rexp</i>	129.34	1.09	879.91	(113.81, 144.86)
<i>sexp</i>	1643.81	327.50	3920.65	(1574.62, 1713.01)
<i>entropy</i>	1.0388	0.5586	1.3546	(1.01, 1.06)
<i>days_to_first_fix</i>	48.78	0.00	202.03	(45.21, 52.34)
<i>exp</i>	4628.11	1205.00	7577.76	(4494.37, 4761.85)

Table A.13

Descriptive statistics of the model variables for Brackets

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	42.13	41.00	26.13	(41.66, 42.61)
<i>ns</i>	1.12	1.00	0.34	(1.11, 1.12)
<i>nf</i>	3.05	1.00	26.25	(2.58, 3.53)
<i>rexp</i>	43.76	1.61	264.21	(38.95, 48.57)
<i>sexp</i>	365.39	143.00	511.38	(356.09, 374.71)
<i>days_to_first_fix</i>	58.59	0.00	214.69	(54.69, 62.50)
<i>entropy</i>	0.4887	0.00	0.7824	(0.47, 0.50)
<i>exp</i>	668.61	246.00	1017.29	(650.10, 687.13)

A.13. Brackets

The heat map of the correlation coefficient among the variables of our interest for the Brackets data set is shown in Fig. A.13.

Table A.13 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

B. Violin Plot

A violin plot combines a box plot with a mirrored kernel density estimate (KDE) of a variable's distribution, where wider sections indicate higher observation density.

We use violin plots throughout this section to compare each variable's distribution across all 13 projects — for example, whether bug resolution times cluster around a particular value or spread widely. Several violin plots are provided in Figs.B.1, B.2, B.3 and B.4, which show density extending into negative values for variables that are non-negative by definition (e.g., *ndev*, *exp*, *days_to_first_fix* cannot be less than zero).

This visualisation artefact arises from kernel density estimation, which uses smooth, unbounded kernels that can extend density estimates beyond the data's true support; it does not indicate negative values in the underlying dataset. All raw values for these variables are greater than 0. Readers should interpret the tails of these plots qualitatively (as indicating low-density regions) rather than literally.

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	42.13	41.00	26.13	Right (0.130)
Camel	47.31	31.00	44.80	Right (1.092)
CoreFX	193.68	188.00	109.81	Right (0.155)
Fabric 8	32.10	31.00	16.72	Right (0.197)
Rails	252.61	232.00	169.90	Right (0.364)
TensorFlow	268.72	285.00	140.67	Left (-0.347)
VSCode	60.60	63.00	27.40	Left (-0.263)
WP-Calypso	138.94	146.00	75.89	Left (-0.279)
Rust	348.79	304.00	270.09	Right (0.498)
Django	77.25	35.00	81.19	Right (1.561)
Nova	168.82	149.00	118.28	Right (0.503)
JGroup	10.83	11.00	7.07	Left (-0.072)
Broadleaf	21.07	24.00	11.00	Left (-0.799)

Table C.1

Summary statistics of *developer count* across all 13 project datasets.

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	0.4887	0.00	0.7824	Right (1.874)
Camel	1.0853	0.8136	1.3254	Right (0.615)
CoreFX	1.17	0.83	1.52	Right (0.671)
Fabric 8	1.0616	0.2108	1.5323	Right (1.666)
Rails	0.6316	0.00	0.8885	Right (2.133)
TensorFlow	0.8850	0.6665	1.0911	Right (0.601)
VSCode	0.7228	0.00	1.0209	Right (2.124)
WP-Calypso	0.77	0.4262	0.99	Right (1.042)
Rust	0.8812	0.2422	1.2184	Right (1.574)
Django	0.7102	0.00	1.0183	Right (2.092)
Nova	0.8764	0.7219	1.0609	Right (0.437)
JGroup	0.3983	0.00	0.8194	Right (1.458)
Broadleaf	1.0388	0.5586	1.3546	Right (1.064)

Table C.2

Summary statistics of *entropy* across all 13 project datasets.

B.1. Violin plot for developer count

B.2. Violin plot for file count

B.3. Violin plot for overall experience

B.4. Violin plot for resolution time

C. Summary statistics

C.1. Summary statistics of developer count across all 13 project datasets

C.2. Summary statistics of entropy across all 13 project datasets

C.3. Summary statistics of *subsystem experience* across all 13 project datasets.

C.4. Summary statistics of *recent experience* across all 13 project datasets.

D. Falsification Summary

Directed Acyclic Graph (DAG) falsification is a causal inference technique used to test whether a given causal model, represented as a DAG, is consistent with a given dataset. In essence, it verifies whether a given causal model, represented as a DAG, can be falsified. This section reports the structural-generalizability test.

Conditional independence (CI) tests form the foundation of falsification using a DAG, as they verify whether or not the above independence assumptions hold. The CI test checks if this condition is satisfied. Conditional independence

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	365.39	143.00	511.38	Right (1.305)
Camel	5149.49	844.00	8191.84	Right (1.577)
CoreFX	3538.41	440.00	8407.69	Right (1.106)
Fabric 8	895.08	179.00	1469.96	Right (1.462)
Rails	428.39	77.0	746.29	Right (1.413)
TensorFlow	20180.28	596.5	34515.36	Right (1.702)
VSCode	4739.20	2312.5	5942.17	Right (1.225)
WP-Calypso	507.74	206.0	778.86	Right (1.162)
Rust	3835.32	673.5	6940.66	Right (1.367)
Django	802.20	268.00	1432.75	Right (1.119)
Nova	363.62	90.0	744.30	Right (1.103)
JGroup	3342.80	2539.0	3167.17	Right (0.761)
Broadleaf	1643.81	327.50	3920.65	Right (1.007)

Table C.3

Summary statistics of *subsystem experience* across all 13 project datasets.

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	43.76	1.61	264.21	Right (0.479)
Camel	63.45	1.06	682.88	Right (0.274)
CoreFX	25.11	1.07	228.37	Right (0.316)
Fabric 8	186.125	1.20	3249.38	Right (0.171)
Rails	65.09	1.119	549.86	Right (0.349)
TensorFlow	17.61	1.13	246.65	Right (0.201)
VSCode	52.53	1.40	422.99	Right (0.363)
WP-Calypso	48.91	1.11	337.92	Right (0.424)
Rust	74.79	1.13	1321.50	Right (0.167)
Django	41.75	1.03	414.70	Right (0.295)
Nova	124.65	1.04	2619.02	Right (0.142)
JGroup	47.87	1.0039	1232.16	Right (0.114)
Broadleaf	129.34	1.09	879.91	Right (0.437)

Table C.4

Summary statistics of *recent experience* across all 13 project datasets.

is assessed using DoWhy's `gcm.falsify` module with a significance level of 0.05 to determine, for each node and each candidate independence statement, whether that statement is violated in the data.

The testing procedure involves the following steps:

1. We form a DAG and consider a dataset.
2. We run a CI test for each node, checking for independence between the node and its non-descendants, conditioned on its parents.
3. We compare these violations to random permutations of the structure and see if the given DAG is significantly better than random.

Now, by permuting node orderings and retesting, we can determine how often random DAGs have fewer or as many LMCs as the original graph. The p-value will indicate whether or not the graph is consistent with the data.

Interpretation of Results: A DAG's structural validity is assessed via the permutation-test p-value, not a raw violation count in isolation: a low p-value means the DAG has fewer violations than nearly all randomly permuted alternatives — e.g., $p = 0.15$ means the DAG outperforms 85% of the permuted graphs — and is not falsified; a high p-value means the DAG performs no better than a random graph and is falsified. We apply this logic via two tests targeting different implied independencies: the **LMC test**, based on each node's Local Markov Condition — independence from non-descendants given its parents — and the stricter **TPa test** (Test of permutation-based d-separation), based on the DAG's full set of implied d-separation statements.

Project	Estimate		Refute		
	Mean	p-value	RCC	DSR	PT
Brackets	0.17132	0.0985	0.17128	0.1667	0.00296
Camel	-0.5296	2.27e-78	-0.5296	-0.5294	0.0050
CoreFX	-0.04294	4.27e-31	-0.04294	-0.04345	0.00063
Fabric 8	-0.5514	1.91e-24	-0.5513	-0.5468	-0.0081
Rails	-0.20324	1.36e-174	-0.20324	-0.20308	-0.00079
TensorFlow	-0.0895	< 0.001	-0.0895	-0.0892	-0.0026
VSCode	-0.0623	5.18e-28	-0.0623	-0.0618	-0.0004
WP-Calypso	-0.29374	1.53e-148	-0.29375	-0.29508	-0.00105

Table E.1

Results of analysis for days per additional developer (ndev) (p-values in bold imply statistical significance. Where the underlying p-value underflows to zero in floating-point computation, it is reported as $< .001$ per convention rather than as an exact 0).

We use the permutation-based falsification procedure implemented in DoWhy's `gcm.falsify` module, which tests these independence implications against random permutations of the DAG's structure. Figures D.1 and D.2 show histograms of LMC violations for permuted DAGs and TPa violations, and each panel's legend shows the colour scheme. The dashed blue and orange lines represent the number of LMC and TPa violations, respectively, in the proposed DAG itself (not a permutation).

We refer to this test as "TPa" throughout, matching the figure legends; TPa is the d-separation-based test described above, not a third, additional test. As expected for a DAG genuinely consistent with the data, the given DAG's violation counts (dashed lines) sit near the lower end of both the LMC and TPa permutation distributions, with the permutation histograms largely overlapping aside from the statistical noise inherent to conditional independence (CI) testing. Because a low LMC/TPa p-value under this permutation test indicates the DAG is structurally superior to almost all randomly permuted alternatives — the opposite of the conventional 'reject at low p' reading — we retain DAGs whose LMC p-value falls at or below a cutoff empirically located at the gap in the observed distribution ($p \approx 0.21$, separating a cluster of retained projects at $p \leq 0.20$ from a cluster of rejected projects at $p \geq 0.22$). This cutoff ($p \approx 0.21$) coincides closely with the `gcm.falsify` module recommended in Eulig [?], which lends independent support to a value that was otherwise located empirically from the gap in our own 13 observed p-values rather than fixed in advance. We report this alignment for transparency, since the cutoff was not pre-registered. Falsification of a DAG does not verify accuracy; instead, it checks whether the DAG has deviated from statistical assumptions. The falsification test shows that the proposed DAG is informative, but it is statistically inconsistent with the observed conditional-independence structure in the remaining five projects.

E. Causal-effect estimates

E.1. Causal effect of ndev on bug resolution time

Table E.1 summarises the causal inference results for developer count on resolution time across different software.

E.2. Causal effect of entropy on bug resolution time

Table E.2 summarises the causal inference results for the effect of *entropy* on resolution time across different software.

Table E.2 shows a large, statistically significant, and refutation-robust estimated effect of entropy on bug resolution time for Brackets, Camel, Rails and WP-Calypso, interpreted as causal under the DAG's identification assumptions. The corresponding estimate is smaller in magnitude, though still statistically significant and refutation-robust, for CoreFX, Fabric 8 and TensorFlow, and smaller still — though still statistically distinguishable from zero — for VSCode.

E.3. Causal effect of sexp on bug resolution time

Table E.3 summarises the causal inference results for subsystem experience on resolution time across software.

E.4. Causal effect of rexp on bug resolution time

Table E.4 summarises the causal inference results for rexp on days_to_first_fix.

Project	Estimate		Refute		
	Mean	p-value	RCC	DSR	PT
Brackets	29.441	2.76e-05	29.4388	29.5595	-0.29752
Camel	19.7228	1.38e-138	19.7227	19.8193	-0.09790
CoreFX	5.18017	1.50e-63	5.18021	5.20880	-0.02837
Fabric 8	5.2500	8.16e-30	5.2498	5.3078	-0.0371
Rails	45.0770	8.33e-188	45.0762	45.1794	0.1222
TensorFlow	8.5550	< 0.001	8.5551	8.5811	-0.0098
VSCode	1.1861	5.64e-36	1.1860	1.1749	-0.0140
WP-Calypso	21.06	1.95e-232	21.0698	21.0578	-0.01190

Table E.2

Results of analysis for days per unit of entropy (p-values in bold imply statistical significance. Where the underlying p-value underflows to zero in floating-point computation, it is reported as $< .001$ per convention rather than as an exact 0).

Project	Estimate		Refute		
	Mean	p-value	RCC	DSR	PT
Brackets	0.01464	0.0951	0.01464	0.01445	0.00020
Camel	0.00111	3.93e-05	0.00111	0.00111	0.00111
CoreFX	8.34e-05	0.011	8.34e-05	7.12e-05	-2.16e-06
Fabric 8	0.00017	0.11	0.00017	0.000201	0.0001
Rails	0.01701	6.94e-23	0.01701	0.01694	-0.00052
TensorFlow	0.00010	3.49e-51	0.00010	0.00010	0.00010
VSCode	-0.00015	0.316	-0.00015	-0.00015	-0.00015
WP-Calypso	0.00358	1.70e-21	0.00358	0.00360	-0.00014

Table E.3

Results of analysis for per unit of *sexp* (p-values in bold imply statistical significance). Where the underlying p-value underflows to zero in floating-point computation, it is reported as $< .001$ per convention rather than as an exact 0.

Project	Estimate		Refute		
	Mean	p-value	RCC	DSR	PT
Brackets	-0.0289	0.58	-0.02901	-0.02790	0.00585
Camel	-0.00092	0.63	-0.00091	0.000306	0.0
CoreFX	0.02380	0.73	0.02379	0.02726	0.00014
Fabric 8	0.00087	0.73	0.00087	0.00220	-4.99e-05
Rails	-0.0049	0.13	-0.00489	-0.00420	0.00068
TensorFlow	0.00203	0.57	0.0020	0.0021	-0.0005
VSCode	0.00016	0.26	0.00016	9.78e-05	0.00013
WP-Calypso	-0.00988	0.86	-0.00987	-0.01004	-0.00018

Table E.4

Results of analysis for per unit of *rexp* (p-values in bold imply statistical significance). Where the underlying p-value underflows to zero in floating-point computation, it is reported as $< .001$ per convention rather than as an exact 0.

Appendix

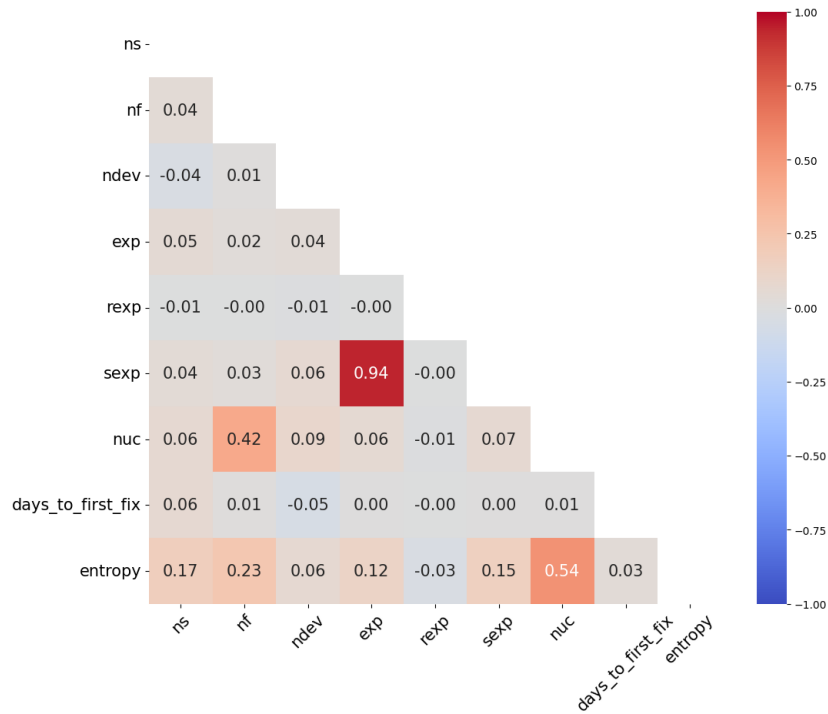


Figure A.1: Heat map of correlation coefficients of Rust dataset

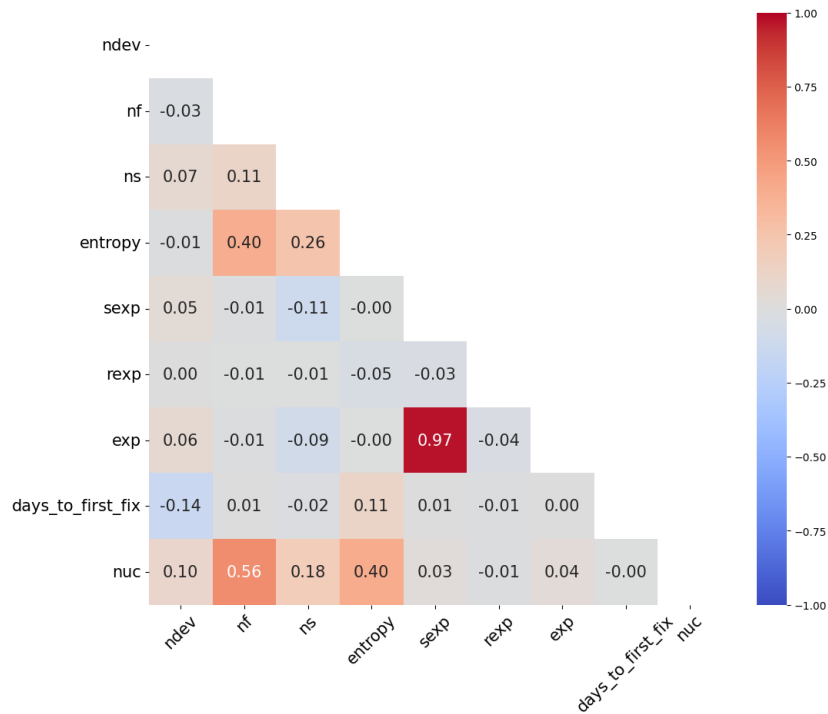


Figure A.2: Heat map of correlation coefficients of Tensorflow dataset

Appendix

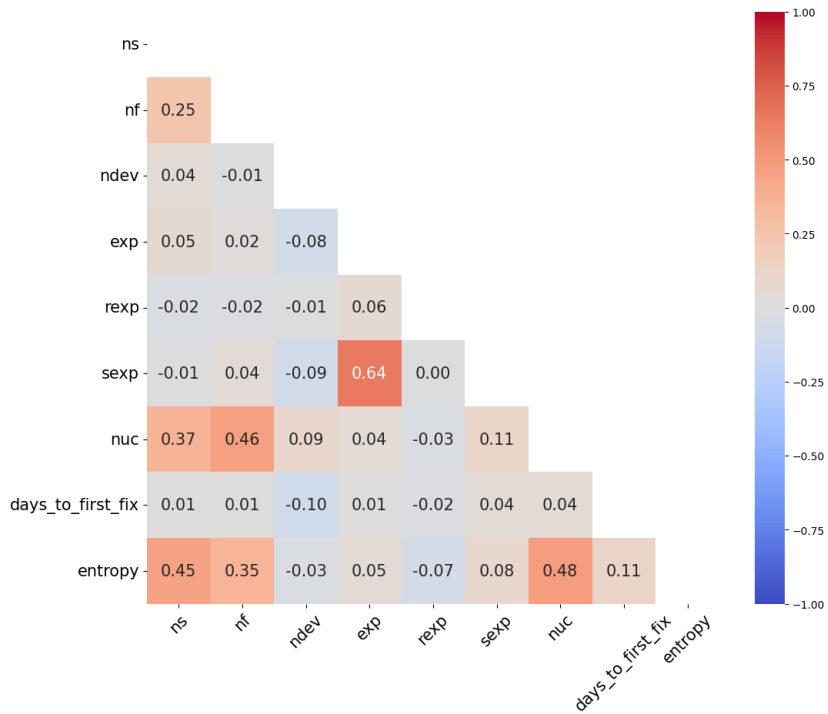


Figure A.3: Heat map of correlation coefficients of Rails dataset

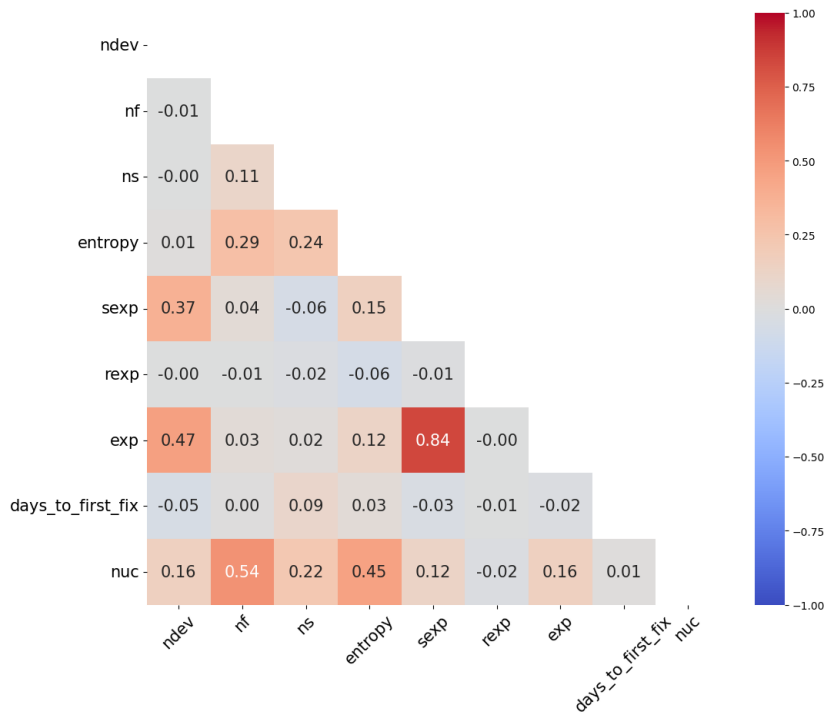


Figure A.4: Heat map of correlation coefficients of VSCode dataset

Appendix

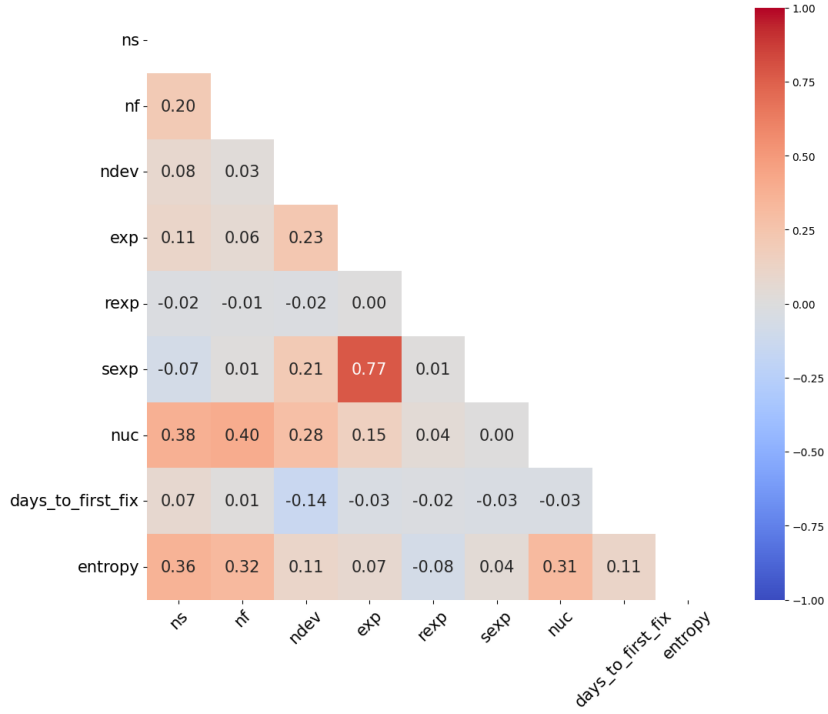


Figure A.5: Heat map of correlation coefficients of WP-Calypso dataset

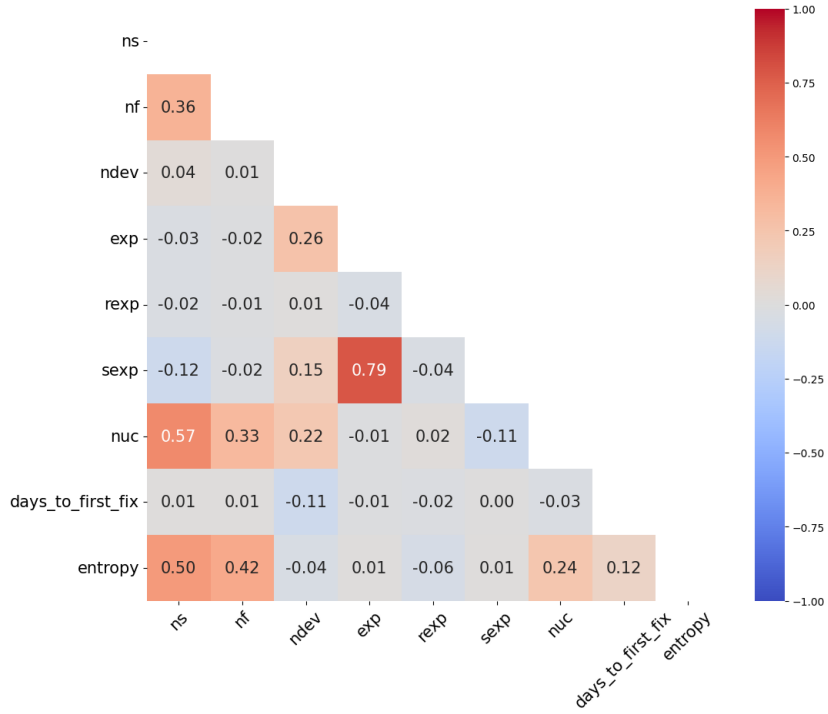


Figure A.6: Heat map of correlation coefficients of Camel dataset

Appendix

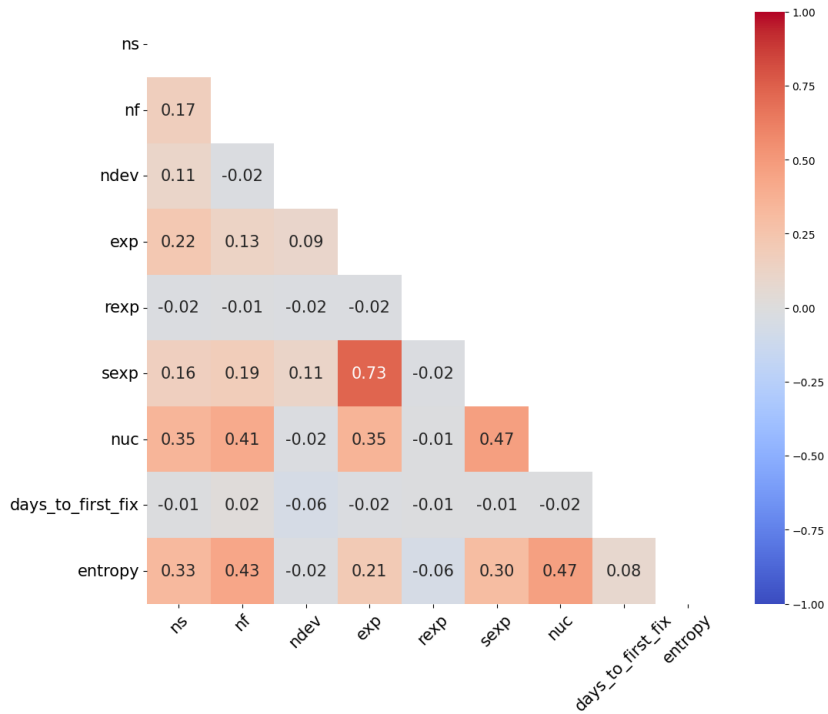


Figure A.7: Heat map of correlation coefficients of CoreFX dataset

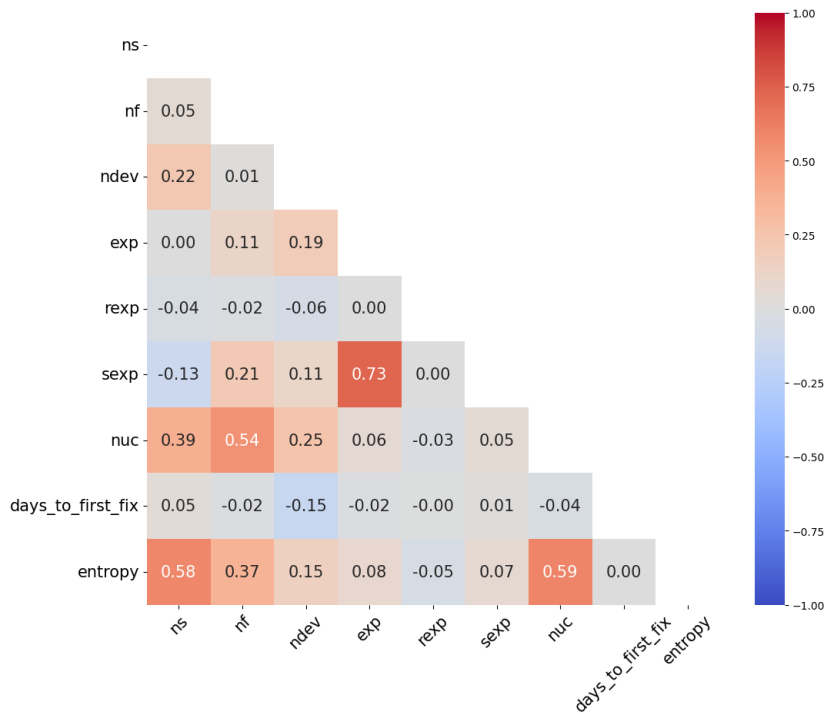


Figure A.8: Heat map of correlation coefficients of Django dataset

Appendix

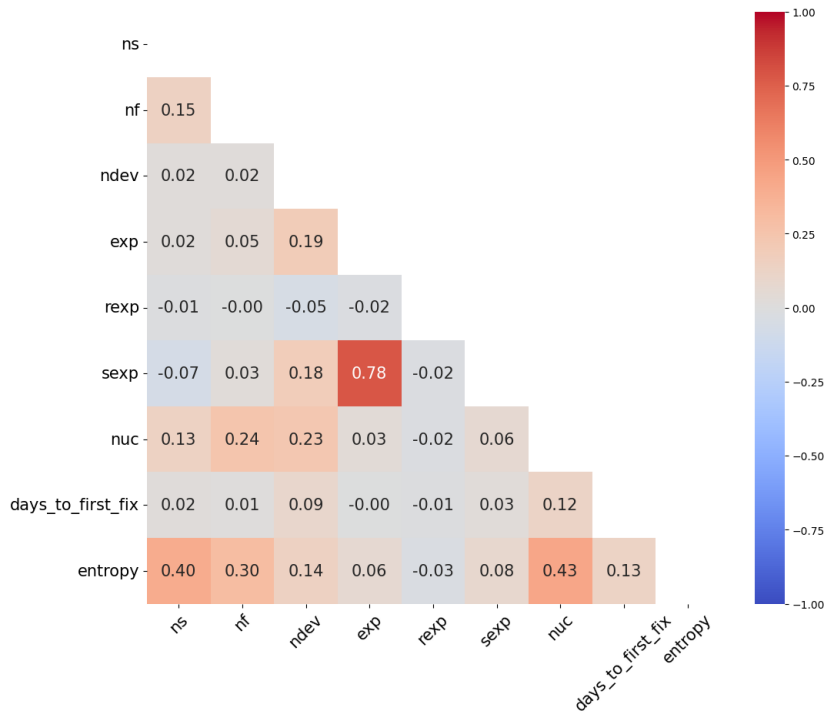


Figure A.9: Heat map of correlation coefficients of Nova dataset

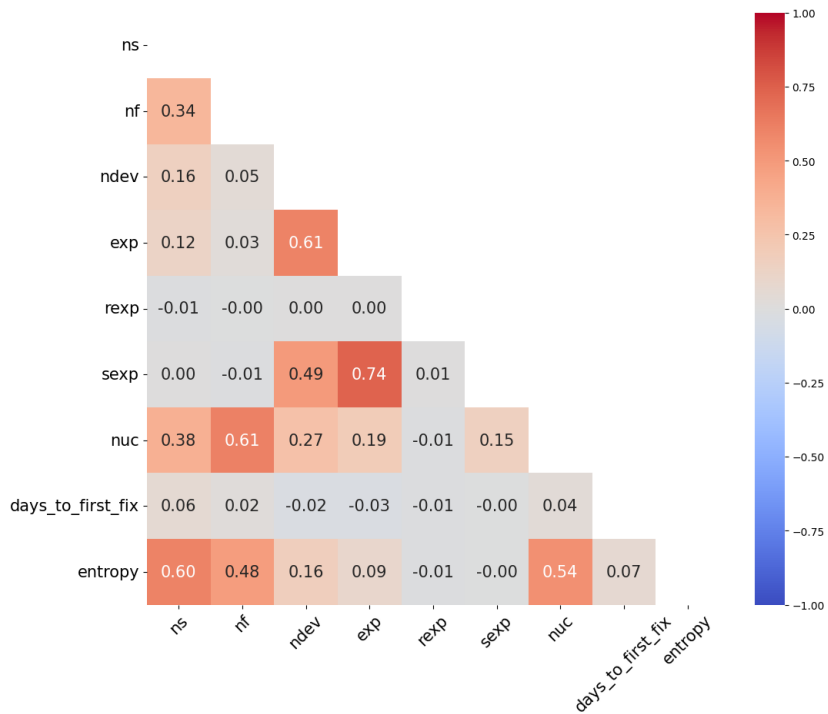


Figure A.10: Heat map of correlation coefficients of JGroup dataset

Appendix

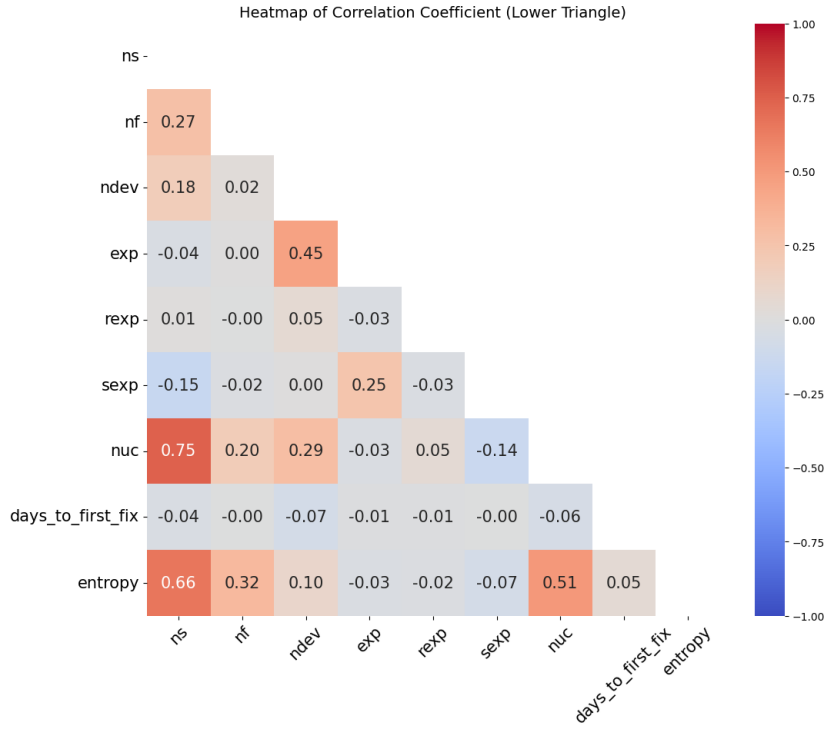


Figure A.11: Heat map of correlation coefficients of Fabric 8 dataset

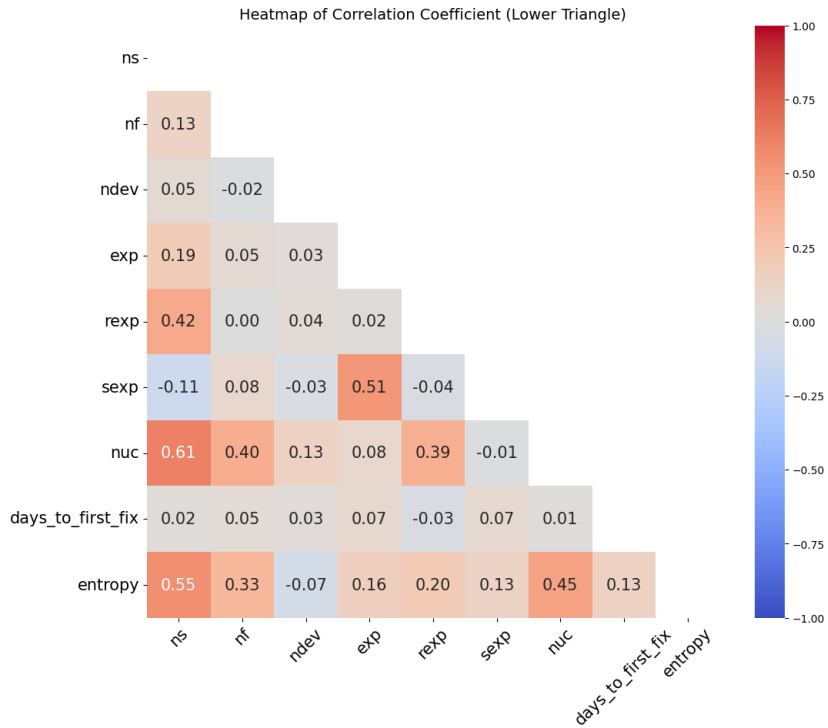


Figure A.12: Heat map of correlation coefficients of Broadleaf dataset

Appendix

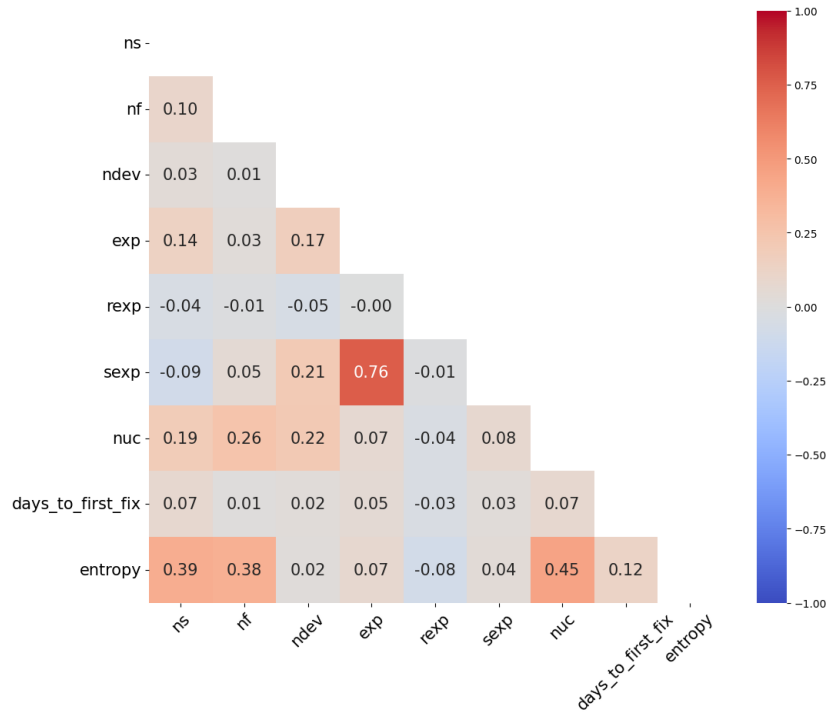


Figure A.13: Heat map of correlation coefficients of Brackets dataset

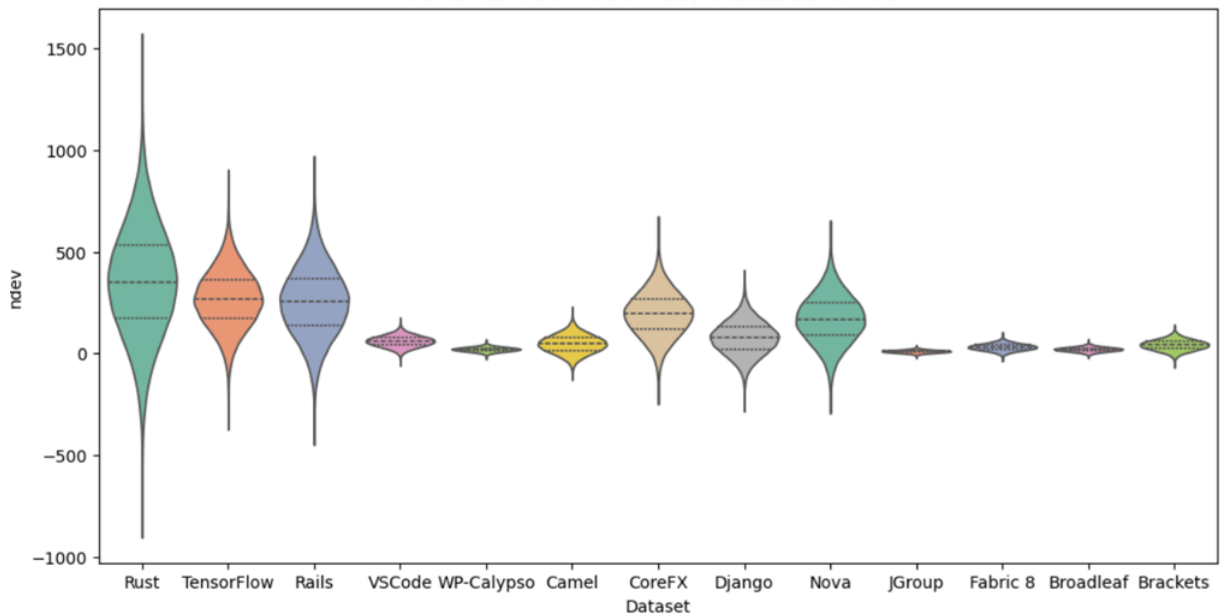


Figure B.1: Violin Plot of Multiple Datasets for ndev

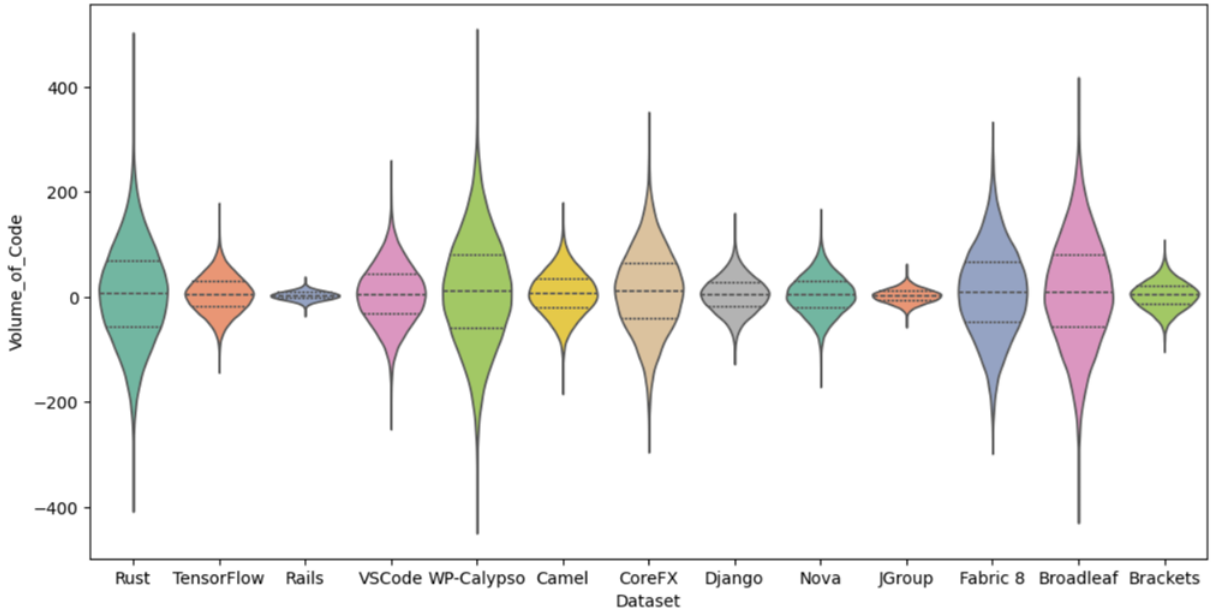


Figure B.2: Violin Plot of Multiple Datasets for Volume_of_Code(= nf)

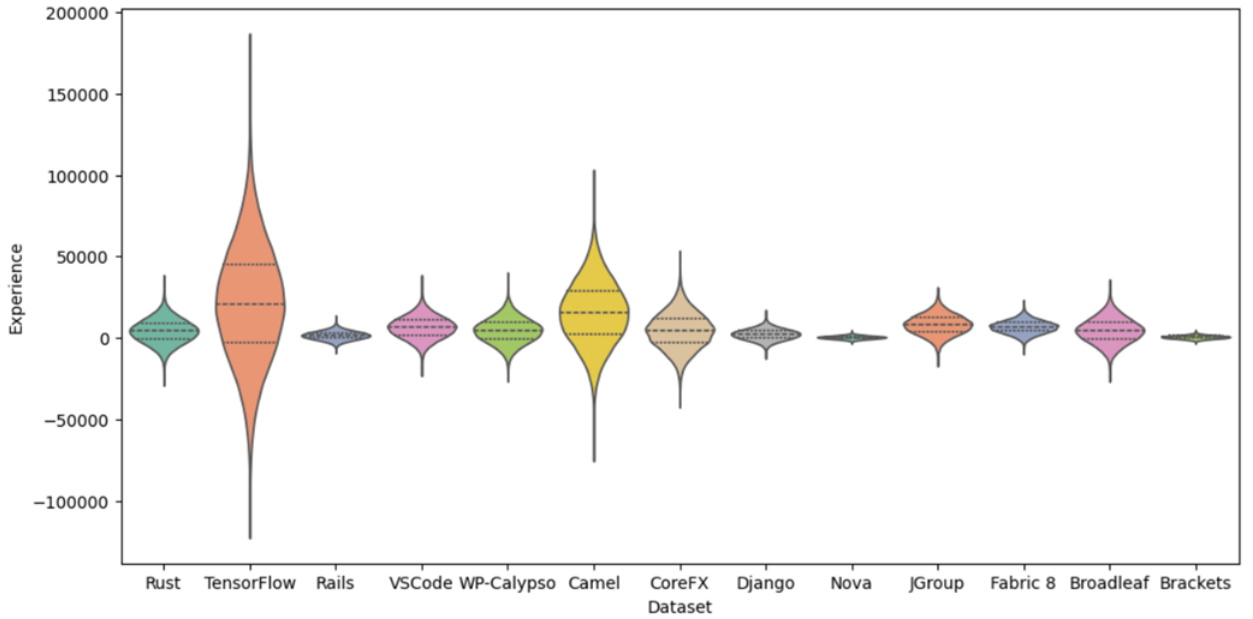


Figure B.3: Violin Plot of Multiple Datasets for Experience (= exp)

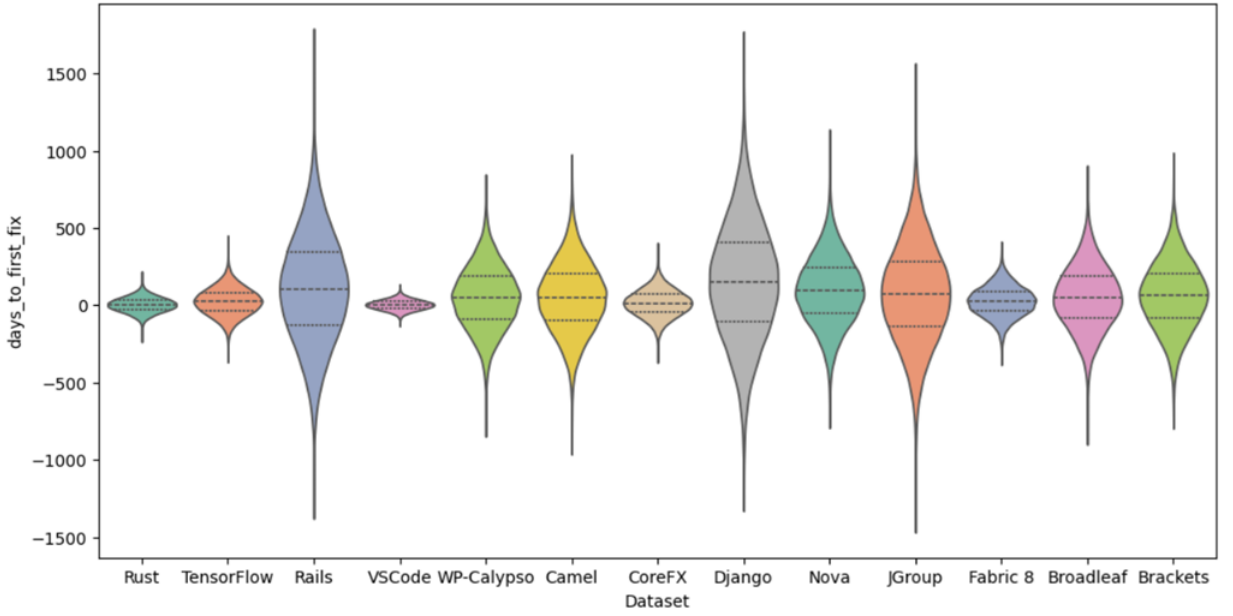


Figure B.4: Violin Plot of Multiple Datasets for Bug Resolution Time

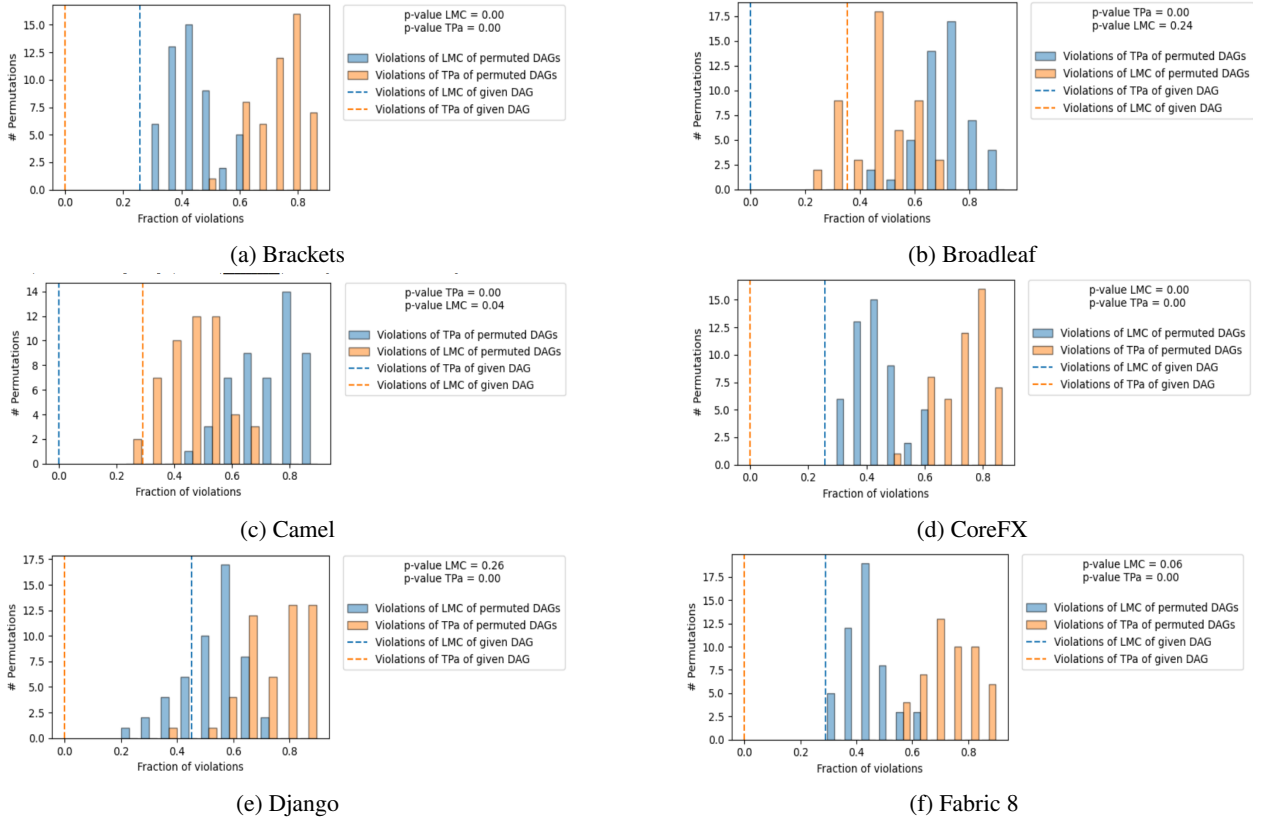


Figure D.1: DAG falsification summary of (a) Brackets, (b) Broadleaf, (c) Camel, (d) CoreFX, (e) Django, and (f) Fabric 8.

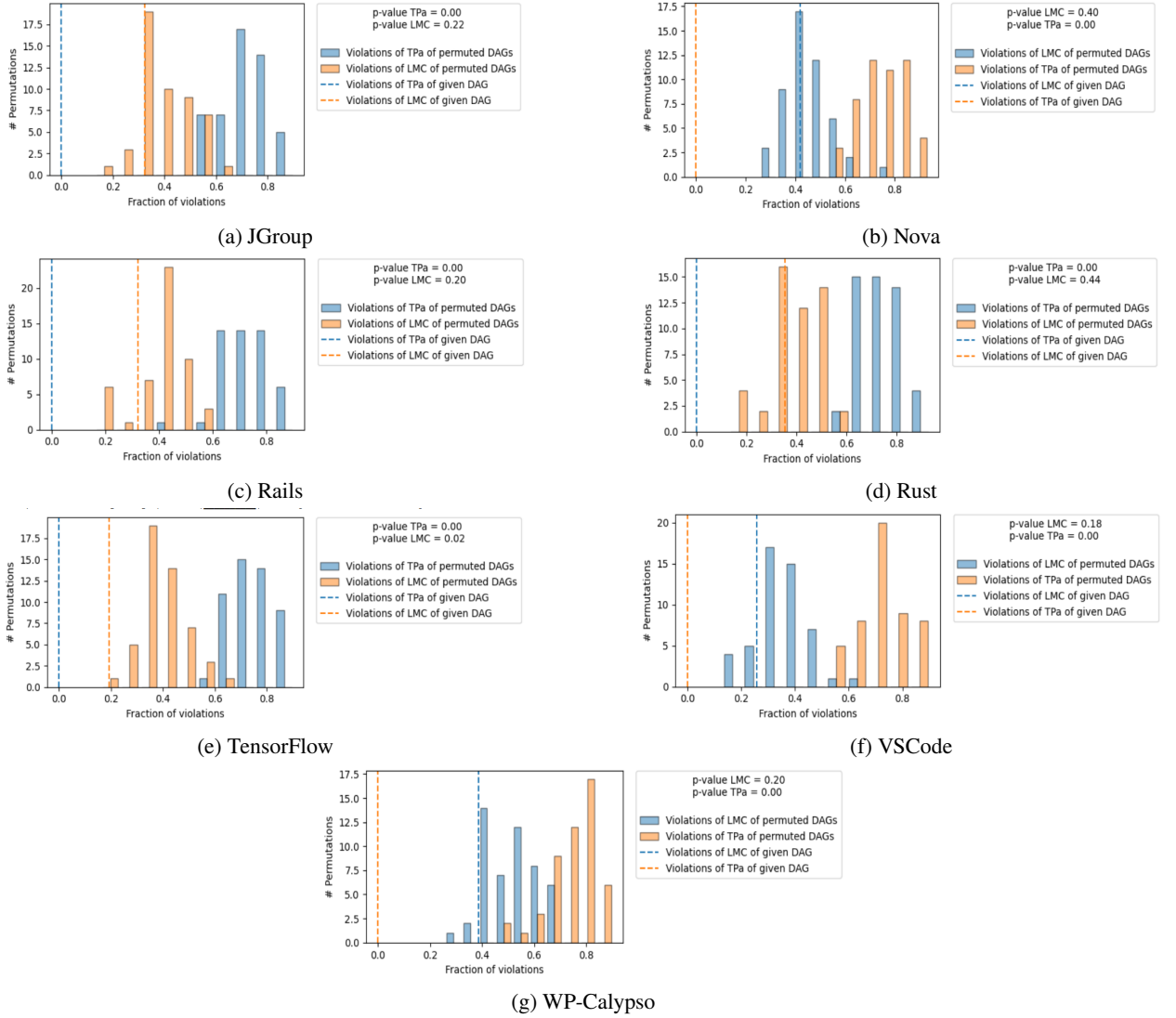


Figure D.2: DAG falsification summary of (a) JGroup, (b) Nova, (c) Rails, (d) Rust, (e) TensorFlow, (f) VSCode and (g) WP-Calypso.